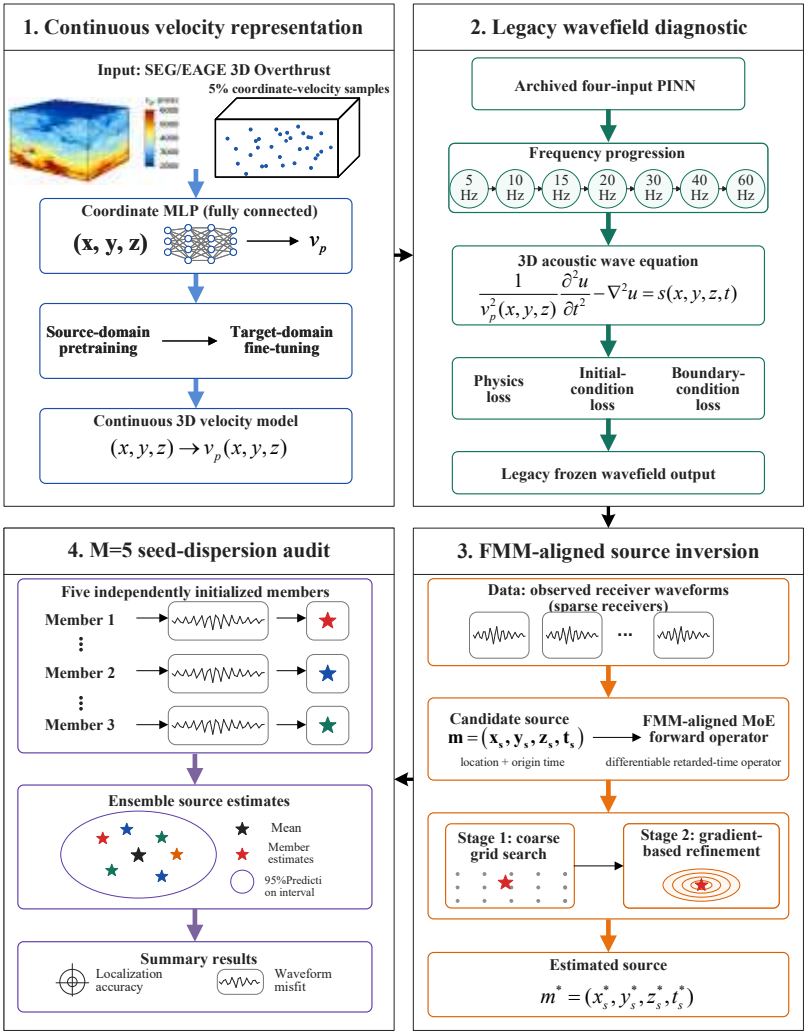

1. Continuous velocity representation
Input: SEG/EAGE 3D Overthrust
5% coordinate-velocity samples
Coordinate MLP (fully connected)
(x, y, z)
v_p
Source-domain pretraining
Target-domain fine-tuning
Continuous 3D velocity model
$(x, y, z) \rightarrow v_p(x, y, z)$
2. Legacy wavefield diagnostic
Archived four-input PINN
Frequency progression
5 Hz
10 Hz
15 Hz
20 Hz
30 Hz
40 Hz
60 Hz
3D acoustic wave equation
$\frac{1}{v_p^2(x,y,z)}\frac{\partial^2 u}{\partial t^2} - \nabla^2 u = s(x, y, z, t)$
Physics loss
Initial-condition loss
Boundary-condition loss
Legacy frozen wavefield output
4. M=5 seed-dispersion audit
Five independently initialized members
Member 1
Member 2
Member 3
Ensemble source estimates
Mean
Member estimates
95%Predicti on interval
Summary results
Localization accuracy
Waveform misfit
3. FMM-aligned source inversion
Data: observed receiver waveforms (sparse receivers)
Candidate source
$\mathbf{m} = (\mathbf{x_s}, \mathbf{y_s}, \mathbf{z_s}, \mathbf{t_s})$
location + origin time
FMM-aligned MoE forward operator
differentiable retarded-time operator
Stage 1: coarse grid search
Stage 2: gradient-based refinement
Estimated source
$m^* = (x_s^*, y_s^*, z_s^*, t_s^*)$